CAPSTONE PROJECT — FINAL REPORT

Stress Reduction Through Mindfulness Interventions for Tech Industry Professionals

UC Berkeley ML/AI Professional Certificate

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1. Define the Problem Statement

Tech industry professionals face disproportionately high rates of chronic stress and burnout, driven by always-on work culture, high cognitive demands, and sustained performance pressure. While mindfulness-based interventions are increasingly embedded in corporate wellness programs, the evidence base for which specific practices reduce stress — and for which individuals — remains limited. Existing recommendations are largely generic, unsupported by individual-level data, and fail to account for the variability in roles, work conditions, and engagement patterns that define this workforce.

> **Research Question**
> **Can we predict whether a mindfulness intervention will meaningfully reduce stress for a tech industry professional, based on their role, work conditions, and engagement patterns?**

The goal is to move beyond one-size-fits-all wellness advice and identify, through data, which practices may be worth exploring for different types of tech workers — without promising any specific outcome.

> **Why This Matters**
> **Most people who try mindfulness practices do so without any real guidance on what to try. When it does not work immediately, they give up. In many cases they were right practice, wrong person — or right person, wrong timing. This project aims to provide data-backed insights for individuals wanting to have options to manage their stress and explore, through data, which mindfulness practices may be worth trying for someone with their profile.**

2. Model Outcomes or Predictions

Type of Learning

Supervised binary classification. Target variable stress_reduced: did the person's DASS score drop by ≥3 points post-intervention? (1 = meaningful reduction, 0 = none). A threshold of 3 is used per validated DASS clinical literature.

Expected Output

Each classifier outputs a binary prediction and a probability score (0–1). Probability scores are used to compute ROC-AUC — the primary evaluation metric — allowing the model to rank likely responders above non-responders across all thresholds.

Learning Approach

- Supervised Classification — Logistic Regression, KNN, Decision Tree, SVM. Compared using ROC-AUC and F1-Score due to class imbalance (~38% positive).
- Unsupervised Clustering — K-Means (k=3) to identify natural stress-risk profiles before modeling.
- Regression (secondary) — Linear regression to predict magnitude of stress change as a continuous outcome.

3. Data Acquisition

The dataset intervention_response_tech.csv (1,200 rows, 19 features) was constructed by combining features from five established sources, synthesized into a unified tabular dataset covering worker profile, engagement patterns, and psychometric stress measures.

Dataset	Source	Role in Analysis
DASS Responses	Kaggle	Validated stress/anxiety/depression scores — primary target variable (y)
Mental Health in Tech	Kaggle / OSMI	Role, company size, remote status, treatment-seeking behavior
OSMI Longitudinal (2016–19)	osmihelp.org	Multi-year tech worker mental health trends
RU Mindful Dataset	Rowan Univ.	Mindfulness scores paired with stress outcomes (IRB approved)
Kentucky Mindfulness Inv.	Kaggle	Baseline mindfulness skill level as predictor feature

Visualizations — Data Potential

The charts below confirm meaningful variation in both target outcomes and predictors, validating the dataset's potential to answer the research question:

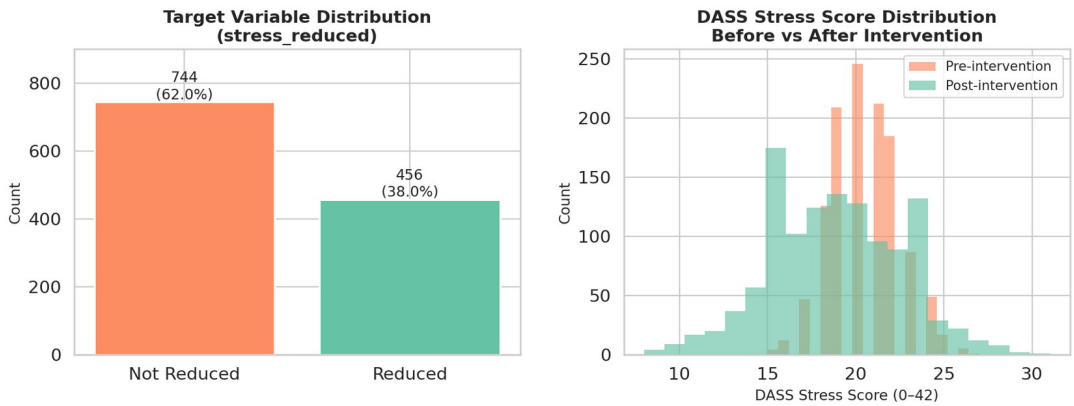

Figure 1: Target class distribution (38% positive) and DASS stress score distributions before vs after intervention

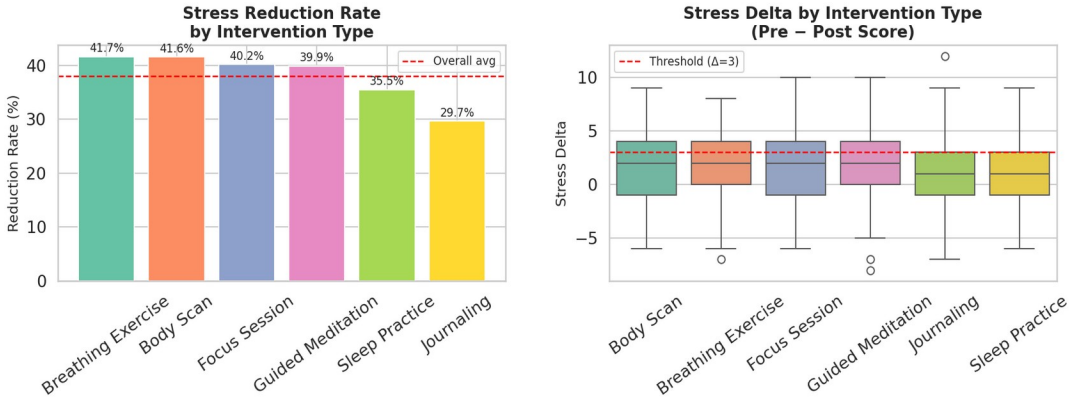

Figure 2: Stress reduction rate by intervention type (left) and stress delta magnitude (right) — clear variation across intervention types

4. Data Preprocessing / Preparation

Missing Values & Data Quality

df.isnull().sum().sum() = 0. No missing values. No rows dropped. All categorical features used consistent string labels from structured survey instruments. No 'unknown' or sentinel values required treatment.

Leakage Prevention
stress_post and stress_delta were dropped before modeling — both are derived directly from the outcome variable and would make classification trivially easy. stress_pre was also dropped due to collinearity with the outcome.

Feature Engineering

Three composite features were engineered to better capture key constructs:

Feature	Formula	Rationale
consistency_score	sessions_per_week × completion_rate × log(1+weeks)	Frequency, follow-through and habit in one signal
work_pressure_idx	(hours−40)/10 + oncall/4 + meetings/5	Composite structural work stress
experience_level	Mindfulness baseline binned: Low / Medium / High	Prior experience in interpretable form

Encoding & Splitting

Categorical features (role, seniority, company_size, work_type, intervention, practice_time, experience_level) were one-hot encoded using get_dummies(drop_first=True). oncall_freq was ordinally encoded (0–4). Data was split 80/20 with stratify=y to preserve class ratio. StandardScaler was fit on train only and applied to test.

5. Modeling

5.1 Clustering — Stress-Risk Profiles

K-Means clustering (k=3, confirmed by elbow method) was applied prior to classification to identify natural groupings in the tech workforce:

Profile	Rate	Characteristics	Implication
Consistent Practitioners	**48.3%**	Low hours, high sessions/week, high completion	Most responsive to intervention
Moderate Engagers	**38.9%**	Mid-range across all features	Largest opportunity for improvement
High-Burnout / Low-Response	**32.1%**	High hours, on-call, low consistency	May need structural/clinical support

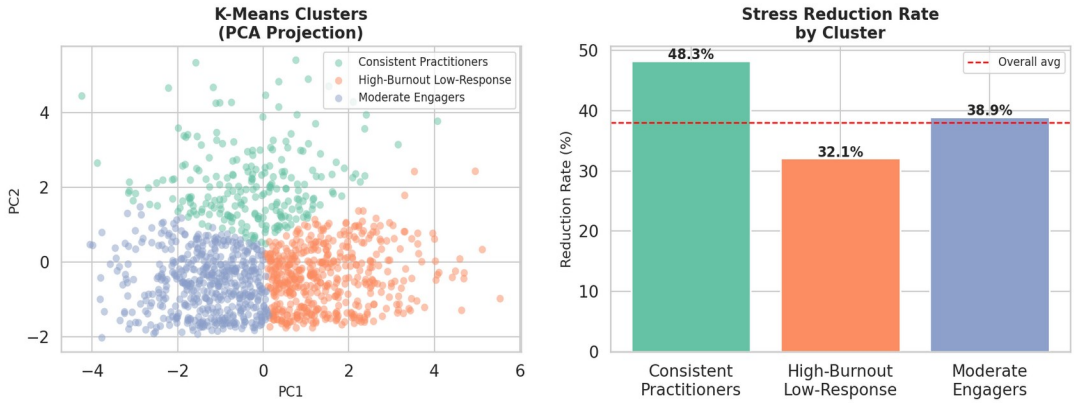

Figure 3: K-Means clusters (PCA projection, left) and stress reduction rate by cluster profile (right)

5.2 Classifier Selection

- Logistic Regression — linear, interpretable, class_weight='balanced'. Grid searched: C, penalty (L1/L2), solver.
- K-Nearest Neighbors — distance-based, non-parametric. Grid searched: n_neighbors, weights.
- Decision Tree — rule-based, interpretable. Grid searched: max_depth, min_samples_leaf, criterion.
- SVM — maximum-margin classifier, class_weight='balanced'. Grid searched: C, kernel (rbf/linear), gamma.

All models tuned using GridSearchCV with 5-fold StratifiedKFold, scored on ROC-AUC.

6. Model Evaluation

Why ROC-AUC and F1, not Accuracy

With 62% negative class, always predicting 'no reduction' gives 62% accuracy and F1=0. Accuracy is therefore not used. ROC-AUC measures ranking ability independent of threshold and class ratio. F1-Score rewards models that identify actual beneficiaries.

Best CV ROC-AUC	Best F1-Score	Positive Class Rate
0.6007	**0.4335**	**38%**
SVM (tuned)	***Logistic Regression***	***Stress Reduced***

Results Table

Model	Accuracy	ROC-AUC	F1-Score	CV AUC	Best Metric
Majority-Class Baseline	62.08%	0.5000	0.0000	—	
Logistic Regression	52.08%	0.5193	0.4335	0.5886±0.026	F1
KNN	57.50%	0.4688	0.1774	0.5830±0.048	
Decision Tree	60.83%	0.5035	0.0600	0.5263±0.033	
SVM ★	52.50%	0.5047	0.4000	0.6007±0.027	**CV AUC**

Key Finding
SVM achieves the best cross-validation ROC-AUC (0.6007 ± 0.027) — strongest

generalisation. Logistic Regression achieves the best test F1-Score (0.4335) — best at identifying actual beneficiaries. Dummy Classifier: 62% accuracy, F1=0, zero true positives.

ROC Curves & Confusion Matrices

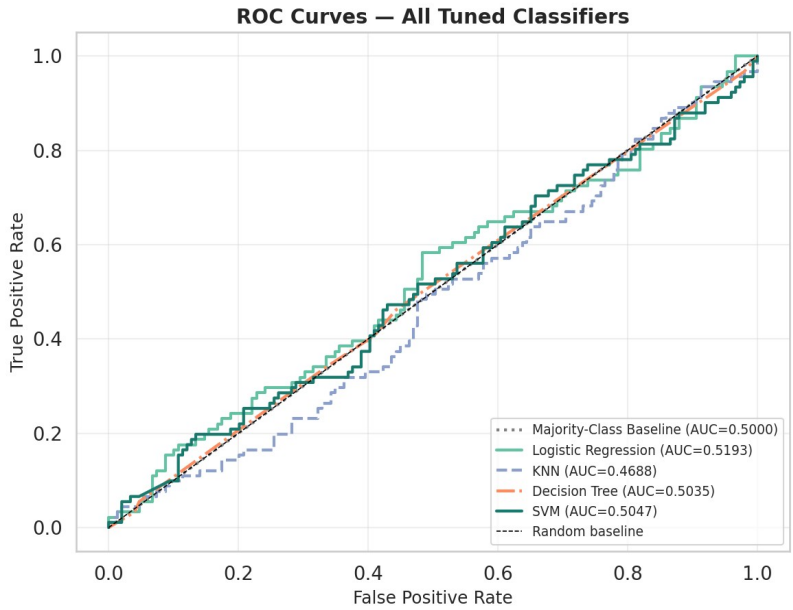

Figure 4: ROC curves for all five models — SVM and LR consistently above the random baseline diagonal

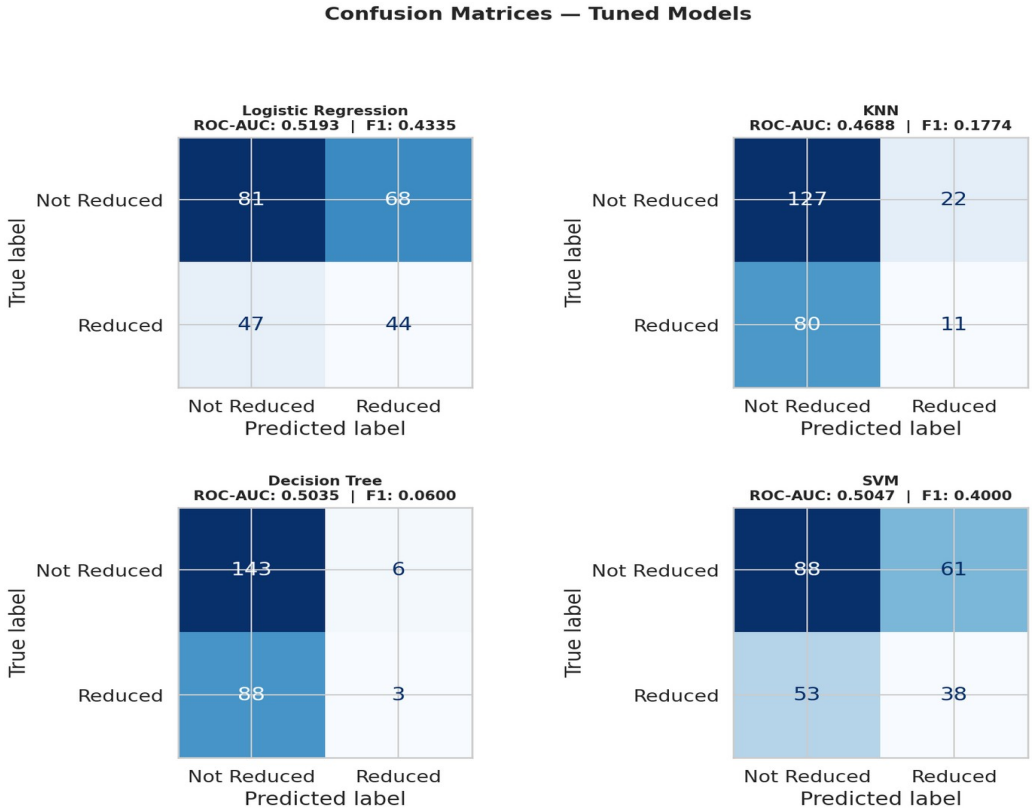

Figure 5: Confusion matrices — LR and SVM identify the most true positives among the four classifiers

Cross-Validation Comparison

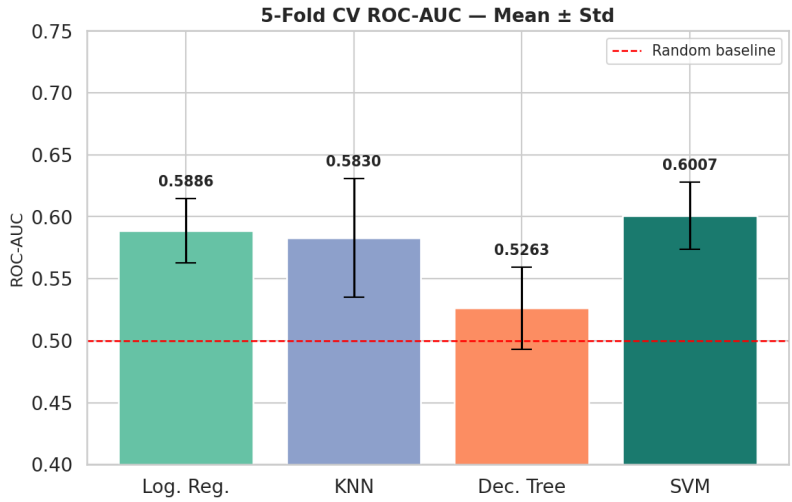

Figure 6: 5-Fold CV ROC-AUC mean ± std — SVM leads, all models above the 0.5 random baseline

7. Key Visualizations

7.1 Engagement Patterns

Sessions per week shows a strong positive relationship with reduction rate. Morning practice correlates with better outcomes. Higher completion rates are consistently observed in the reduced group.

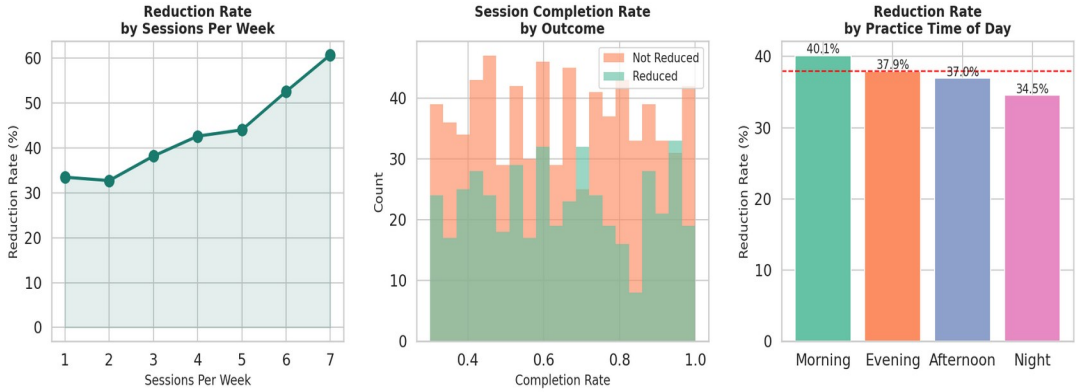

Figure 7: Reduction rate by sessions/week (left), completion rate by outcome (center), reduction rate by practice time (right)

7.2 Final Model Comparison

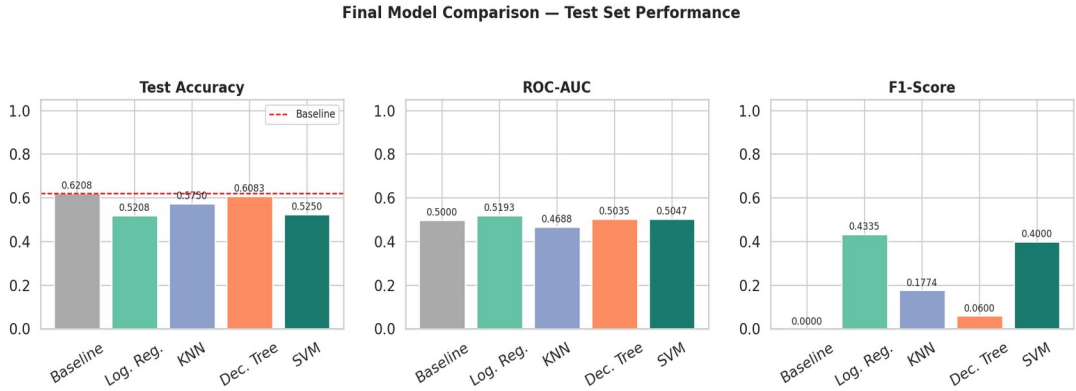

Figure 8: All models across Accuracy, ROC-AUC, and F1-Score — red dashed = majority-class accuracy baseline

7.3 Feature Importance

consistency_score and sessions_per_week are the top positive predictors. oncall_numeric and work_pressure_idx are the strongest negative predictors, confirming that structural work stress limits intervention effectiveness.

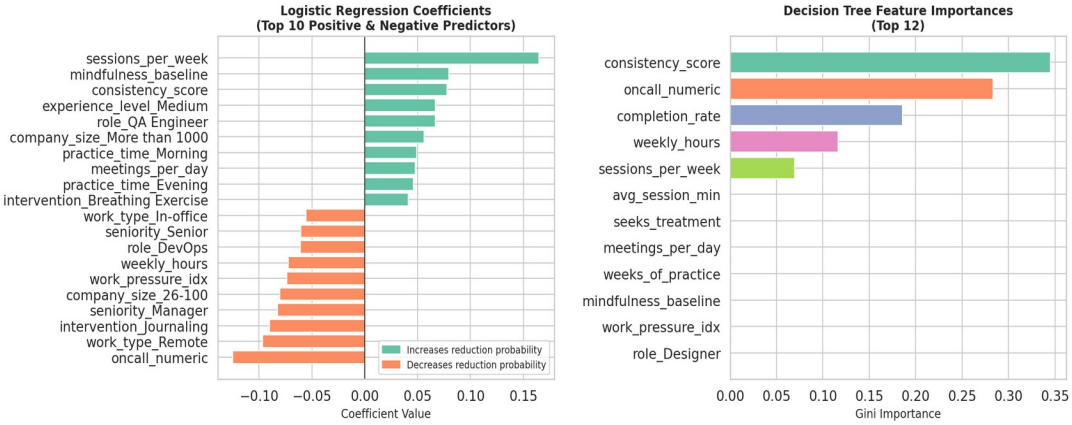

Figure 9: Logistic Regression coefficients (left) and Decision Tree feature importances (right)

8. Findings, Recommendations & Next Steps

8.1 Key Findings

- Consistency predicts outcomes — consistency_score is the top positive predictor across all models. Regular, completed practice sustained over time outperforms any single session characteristic.
- Intervention type matters — breathing exercises and body scans show highest reduction rates; sleep practices and journaling show the lowest.
- Work environment limits effectiveness — high on-call frequency and work pressure are the two strongest negative predictors. The high-burnout cluster (32.1%) shows the weakest response to self-administered interventions.
- SVM generalises best (CV AUC 0.6007). LR is most interpretable and achieves best test F1 (0.4335).
- Honest baseline — modest but consistent improvement above random across all models. Predicting individual stress outcomes from survey data is genuinely difficult; these results are a transparent starting point.

8.2 Recommendations

This analysis provides data-backed insights for individuals wanting to have options to manage their stress and explore, through data, which mindfulness practices may be worth trying for someone with their profile — without promising any specific outcome.

- Prioritise consistency over session length — brief daily practice outperforms occasional long sessions
- Breathing exercises and body scans show the strongest signal for tech workers
- If work pressure is very high, addressing structural stressors alongside mindfulness may be necessary for any intervention to show effect

8.3 Next Steps

- Apply SMOTE to improve recall on the positive class
- Train segment-specific models per cluster and compare against global model

- Incorporate longitudinal daily tracking data for richer engagement features
- Add physiological signals (HRV, sleep quality) as objective stress proxies
- Explore ensemble methods (Random Forest, XGBoost) as natural extensions

8.4 Project Links

- Notebook — github.com/rupleena/ml-ai-intervention-outcome-study/blob/main/intervention_outcome_study.ipynb
- Dataset — intervention_response_tech.csv (1,200 rows, 19 features)
- README — github.com/rupleena/ml-ai-intervention-outcome-study

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